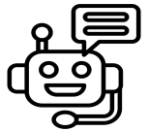
Workflow Development with n8n

Class two

Types of AI-Powered Solutions

- Workflow 
  - Automated **process pipeline**
- Intelligent Chatbot 
  - The system could perform **iterative queries and analysis** based on the end user queries
- Agentic Assistant 
  - It will understand, plan and execute the tasks **autonomously**



PDF Summarization Engine

Resume Summarization engine

Types of nodes in n8n-1/2

Category	Purpose	Examples of Nodes
Trigger Nodes	Start a workflow when something happens	Webhook Trigger, Schedule Trigger (Cron), Gmail Trigger (new email), Telegram Trigger (new message)
App / Action Nodes	Perform operations with specific apps & services	Google Sheets (add row), Gmail (send email), Slack (post message), Notion (create page), Airtable (update record)
Data Transformation	Clean, reshape, or restructure data	Set, Move/Remove Fields, Rename Keys, Item Lists, Split In Batches
Logic / Control Flow	Control workflow execution path	IF, Switch, Merge, Wait, Error Trigger
Function / Code	Add custom programming logic	Function (JavaScript), Function Item (per item JS), Code (JavaScript / Python)
AI & LangChain	AI reasoning & tool calling	OpenAI, Azure OpenAI, AI Agent (LangChain), Think Node, Embeddings
Utility Nodes	General-purpose helpers	HTTP Request, Webhook, Date & Time, Crypto, Spreadsheet File, File Operations
Database Nodes	Read/write structured data	MySQL, PostgreSQL, MongoDB, SQLite, Notion DB, Airtable
Messaging & Notifications	Communicate with users	Email (IMAP/SMTP/Gmail/Outlook), Slack, Teams, Telegram, Discord, Twilio (SMS/WhatsApp)

Types of nodes in n8n-2/2

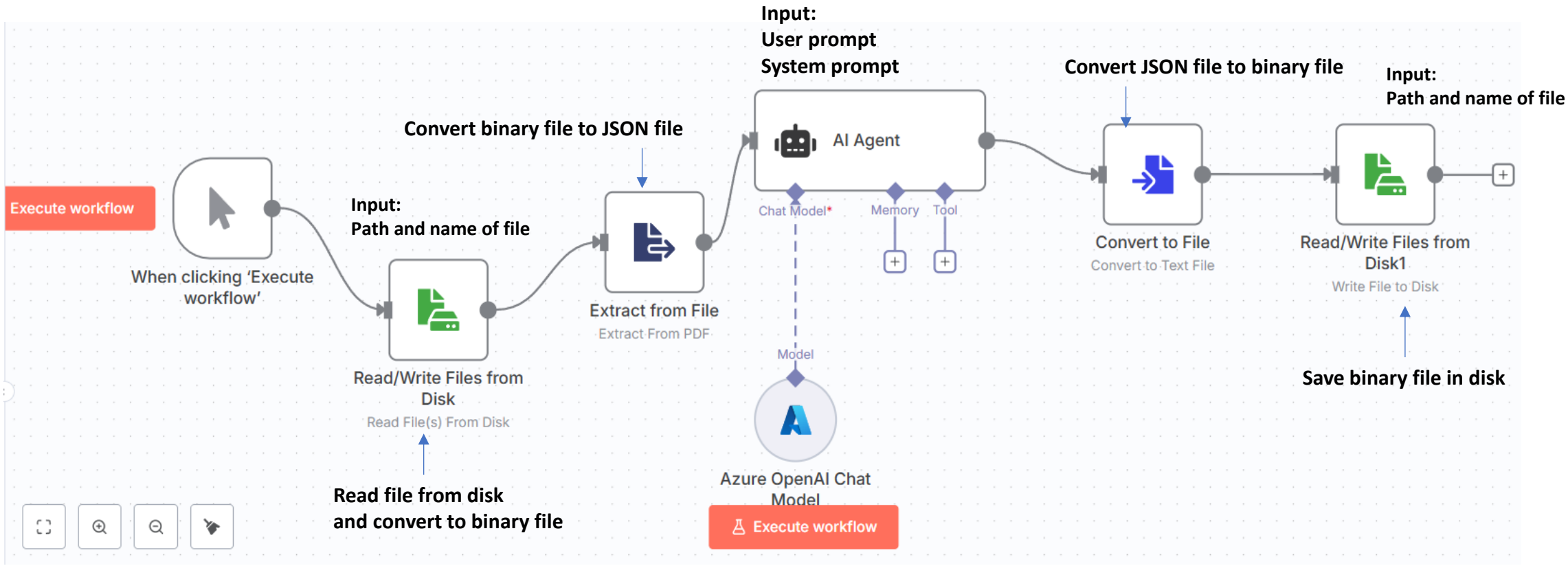
Node	
Read/Write files from Disk	Import files(e.g. pdf/csv) to/from binary
Extract from file	Convert binary file to JSON
Convert to file	Convert JSON data to binary file
Edit Fields	Create/edit content (JSON)
AI Agent	AI framework
Memory	Memory module used in AI framework
LLM	e.g. Azure and other LLM
Web Hook	To receive http requests from other server
Command Node	To execute command (e.g. execute bat file under Windows OS)
HTTP Request	To send HTTP request to other nodes/server/AI Agent
Wait	To pend and wait for further message (e.g. web hook) or condition (time expired)
IF/SWITCH	Condition branching
Application nodes	Google/Meta/Telegram/Whatsapp/etc



Case one

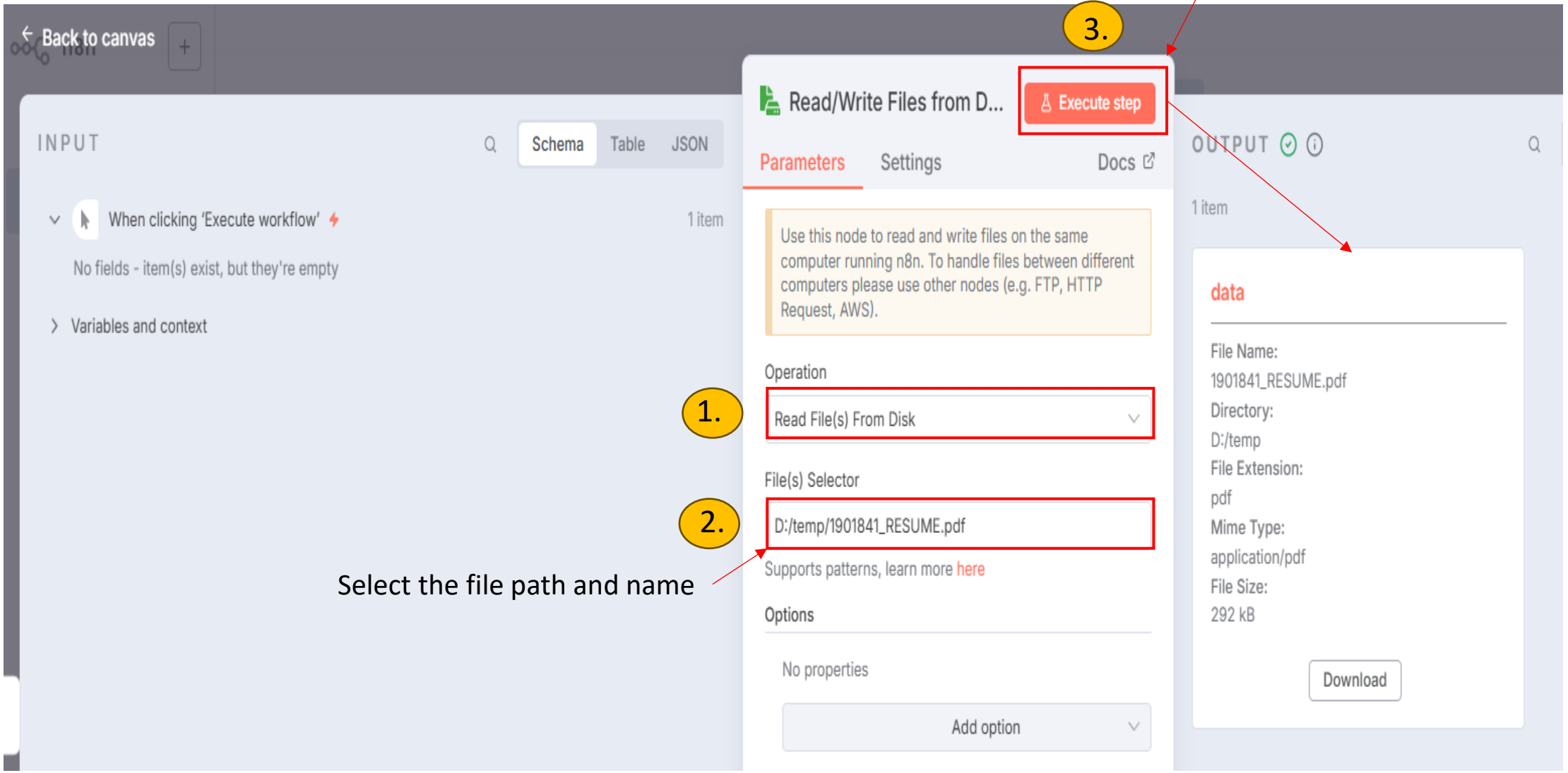
Single PDF file

Use Case
Resume summarization and analysis engine



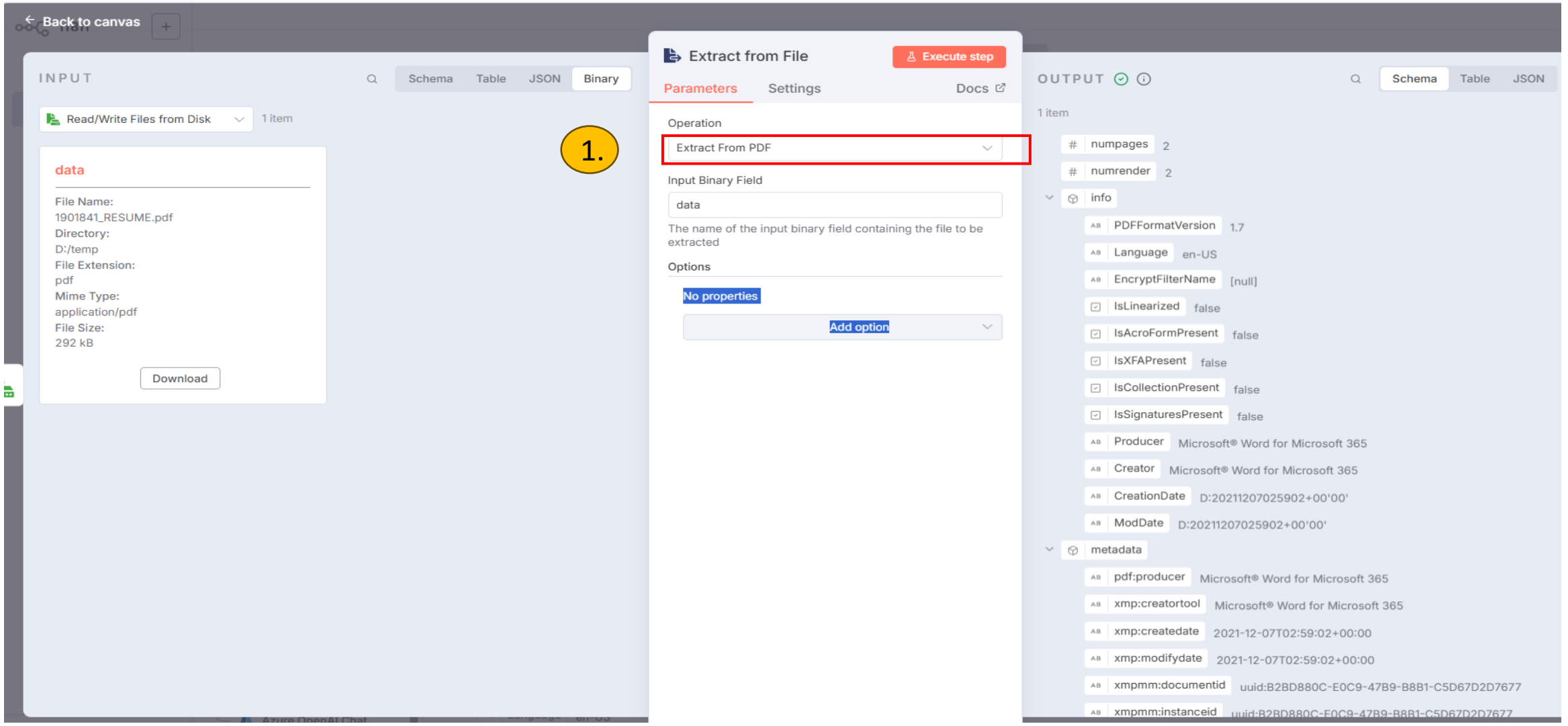
1. Read/Write Files from Disk

Click this one and see the execution result



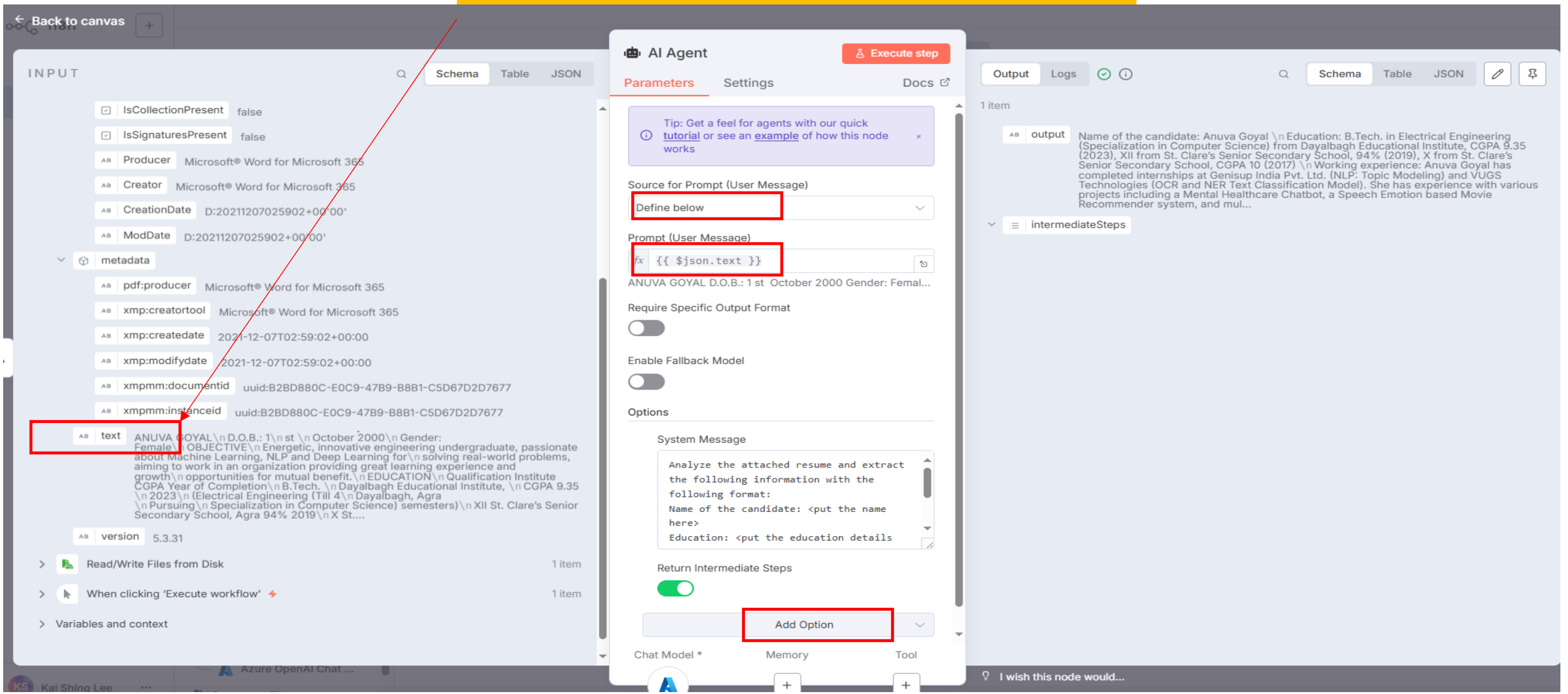
Select the file path and name

2. Extract from file



3. AI Agent

You could simply drag and drop from here to include the variable {{$json.text}}



3. AI Agent
System prompt

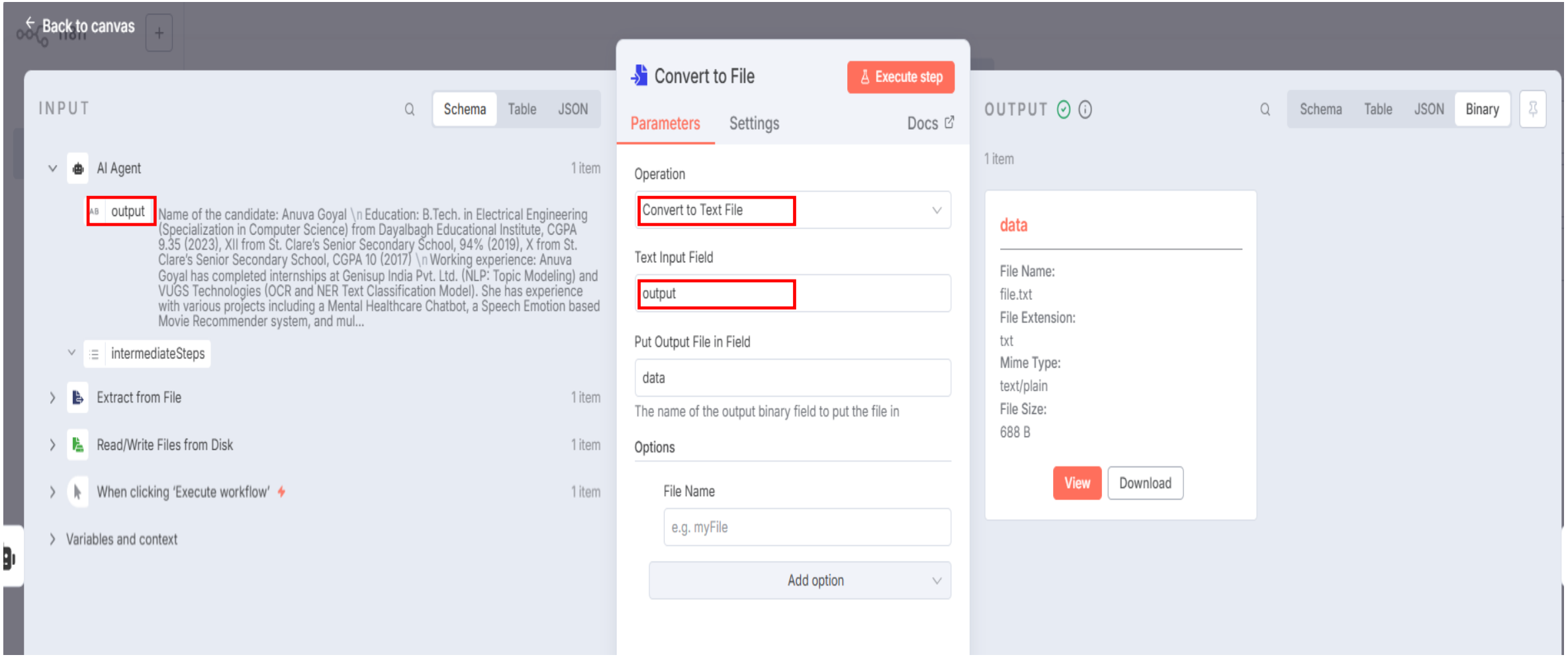
Analyze the attached resume and extract the following information with the following format:

Name of the candidate: <put the name here>

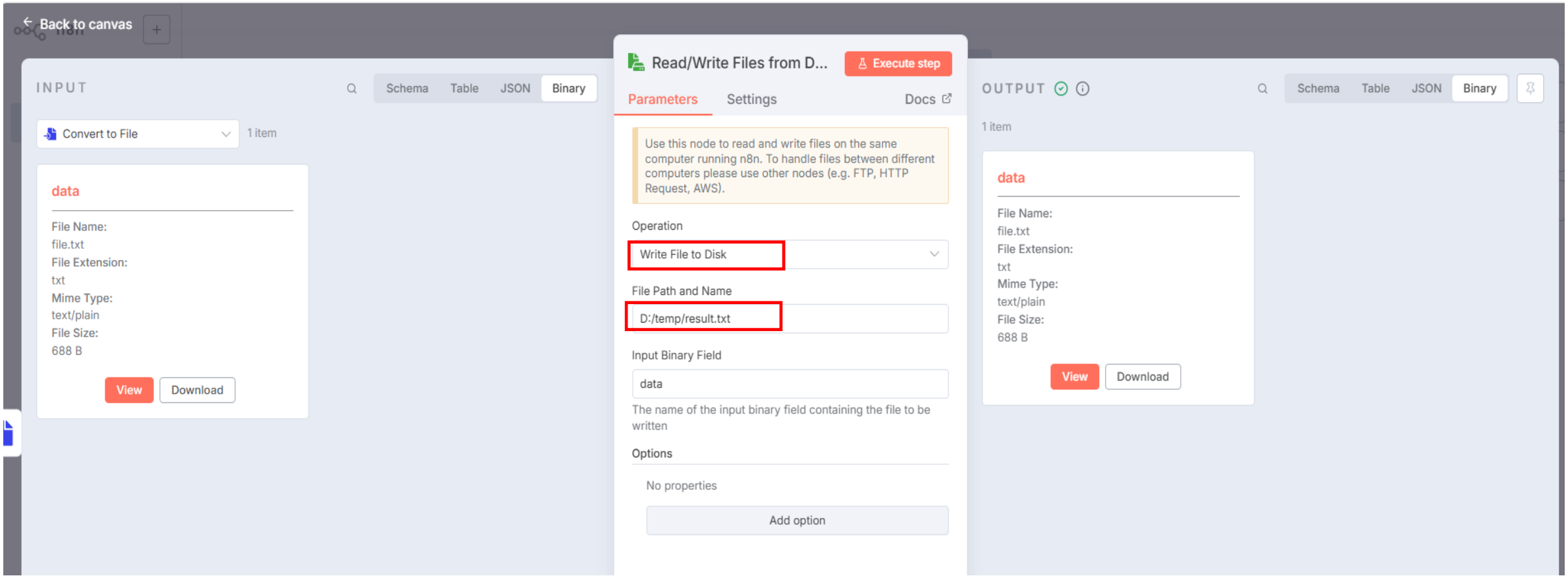
Education: <put the education details here>

Working experience: <put the summarize of the working experience>

4. Convert to file



5. Read/Write Files from Disk

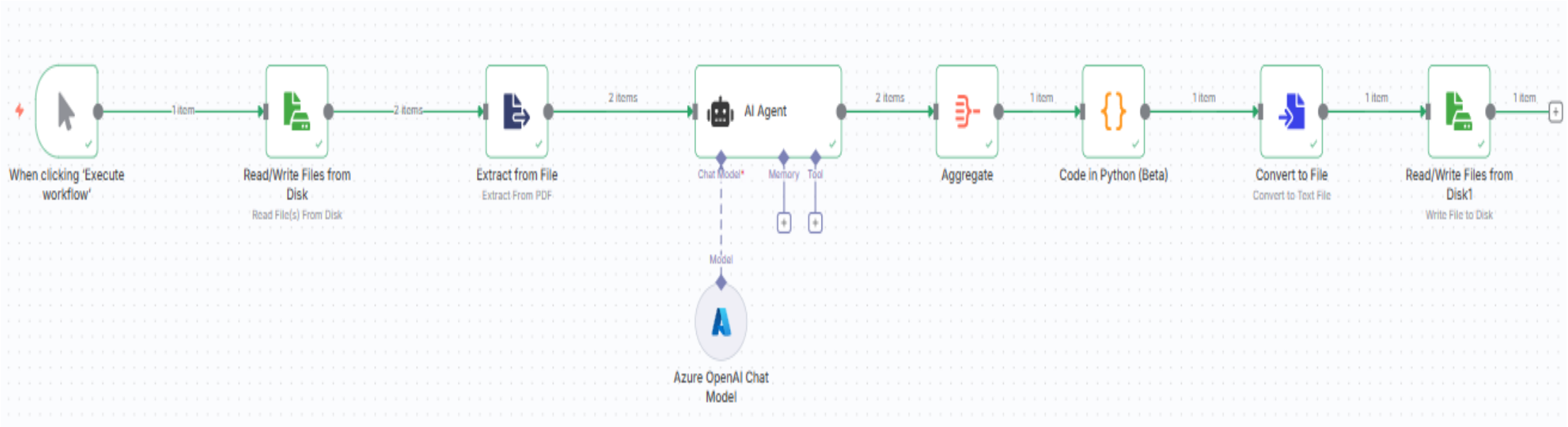




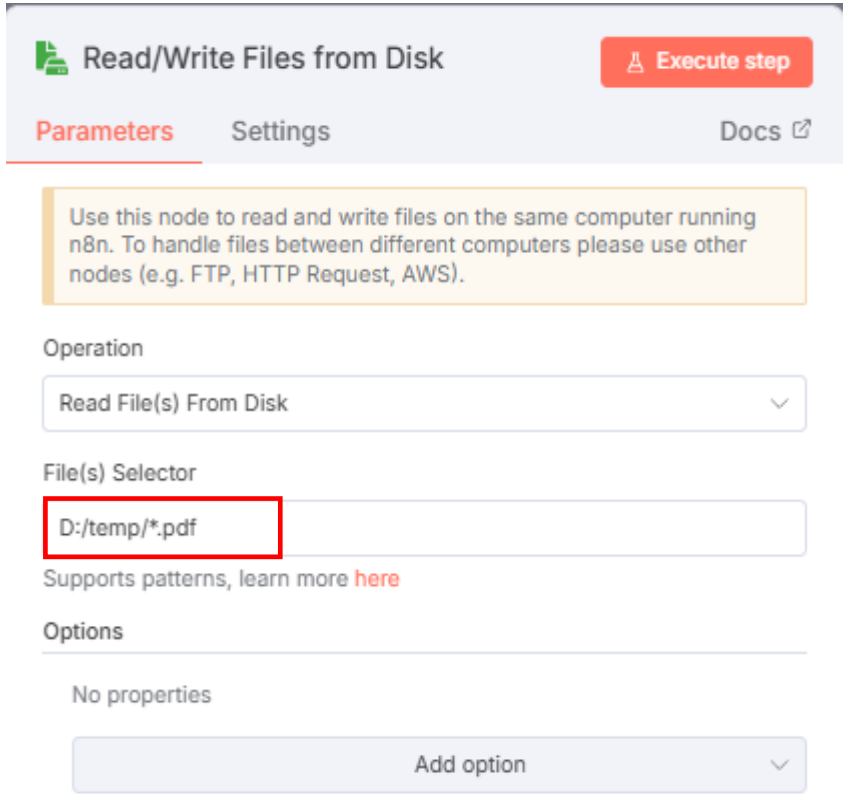
Case two

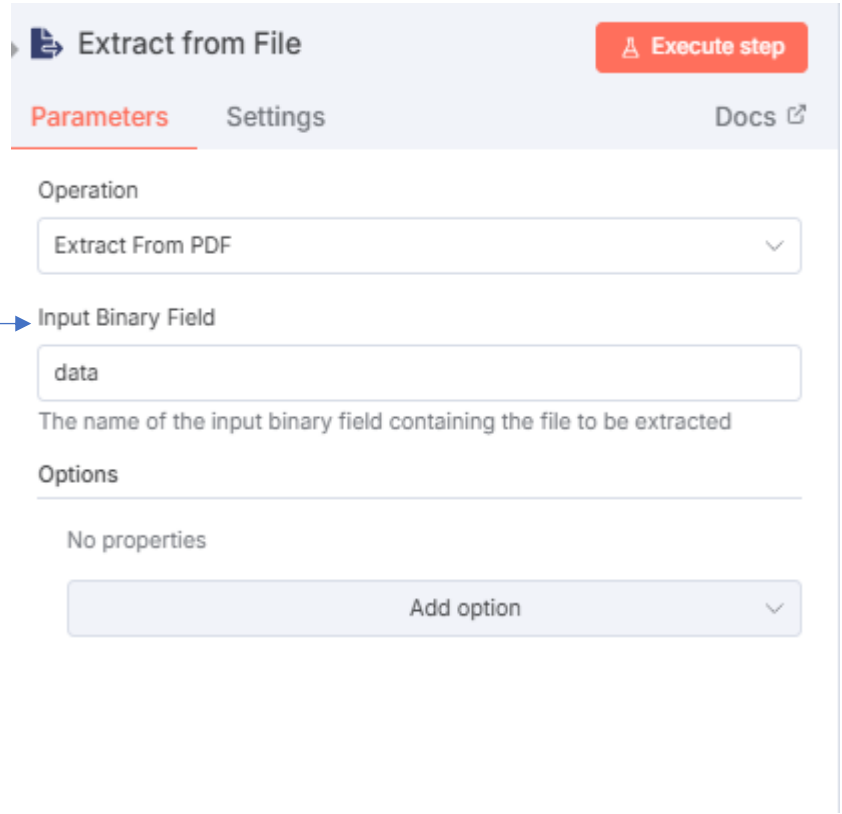
Multiple pdf files

Resume Summarization (Multiple files)

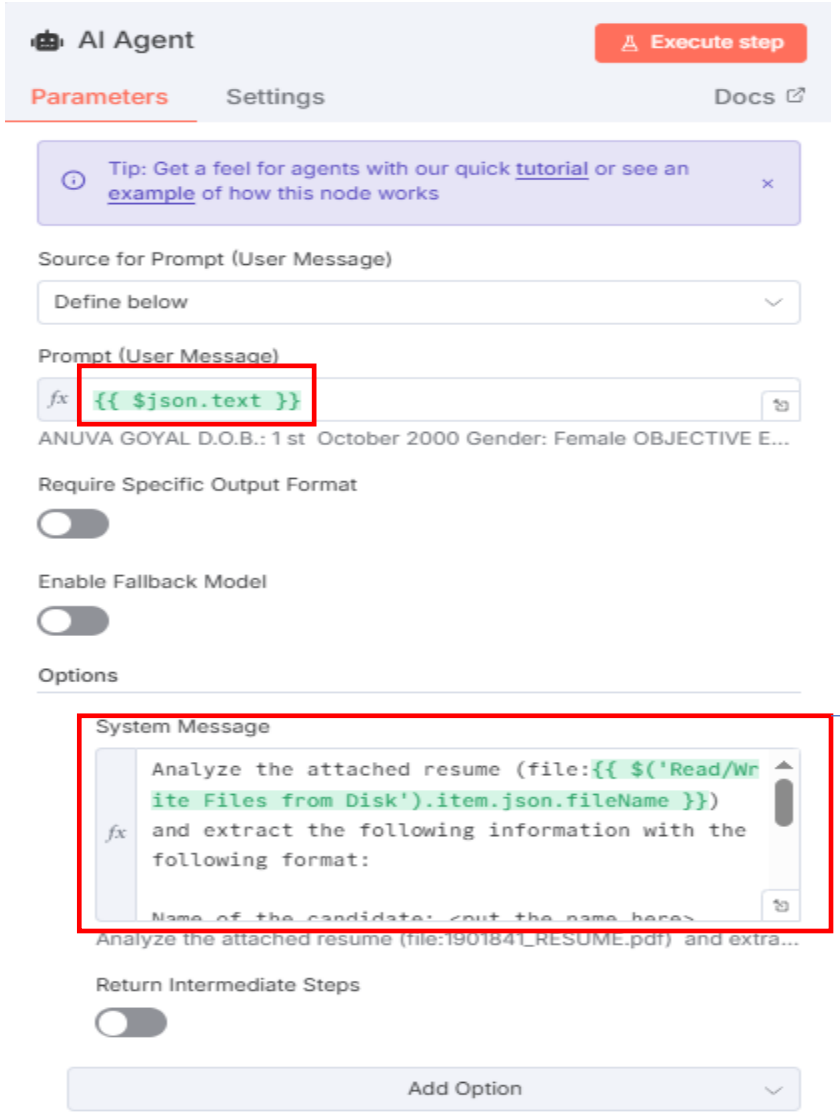


Read/Write Files from Disk & Extract from File





AI Agent



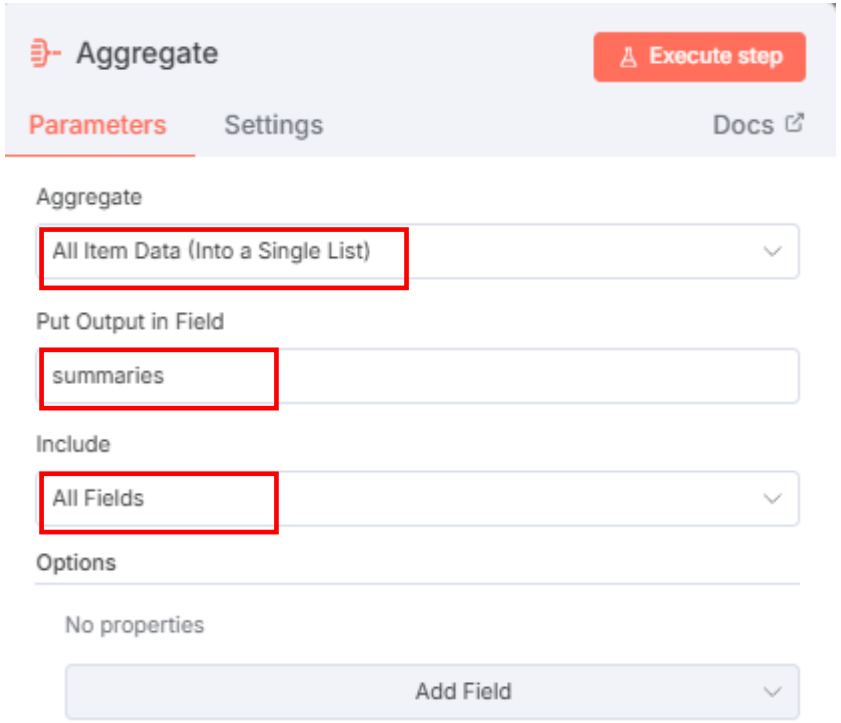
System Prompt

```
Analyze the attached resume (file:{{ $('Read/Write Files from Disk').item.json.fileName }})
and extract the following information with the following format:

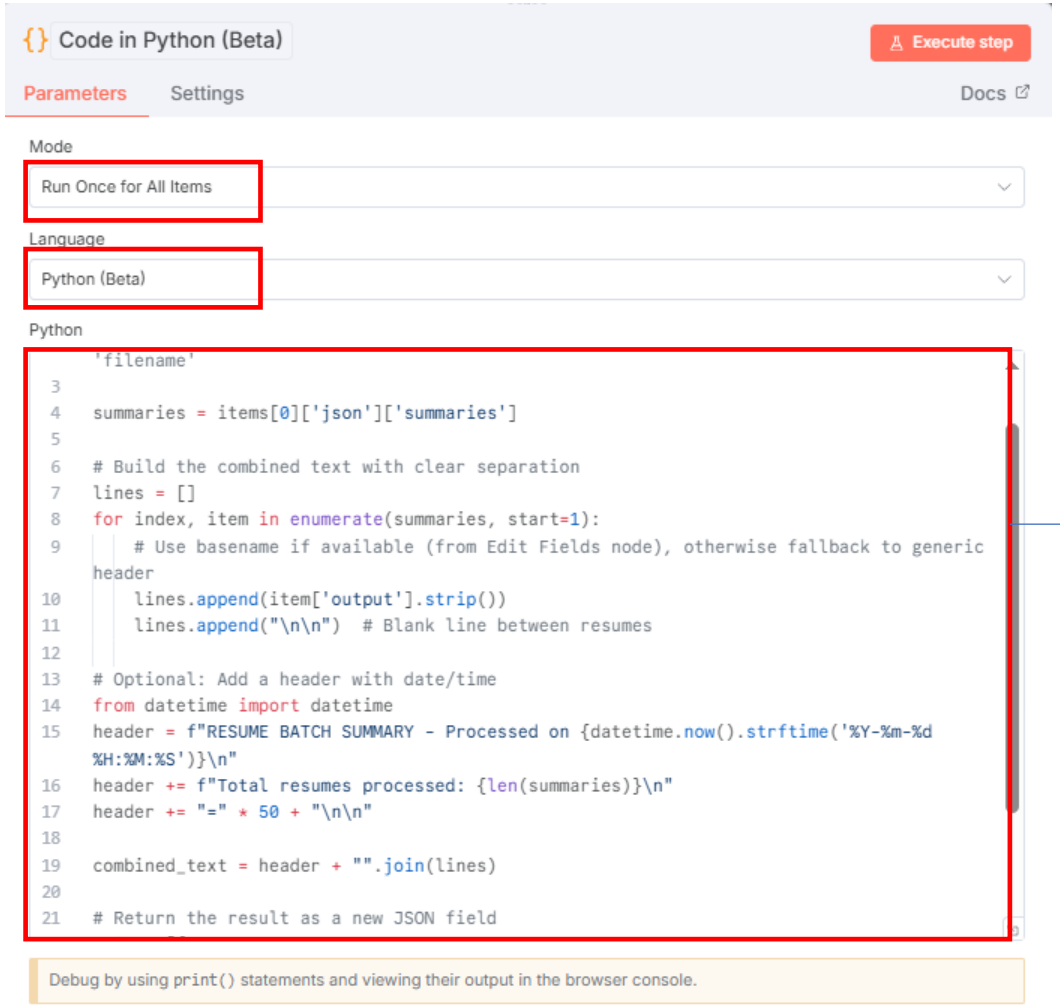
Name of the candidate: <put the name here>
Education: <put the education details here>
Working experience: <put the summarize of the working experience>

Start your response with:
File: {{ $('Read/Write Files from Disk').item.json.fileName }}
```

Aggregate Node



Code Node



Python Code

```
# Input: items[0].json.summaries is a list of dicts from the Aggregate node
# Each dict contains at least 'output' (the AI summary), and possibly 'basename' or 'filename'

summaries = items[0]['json']['summaries']

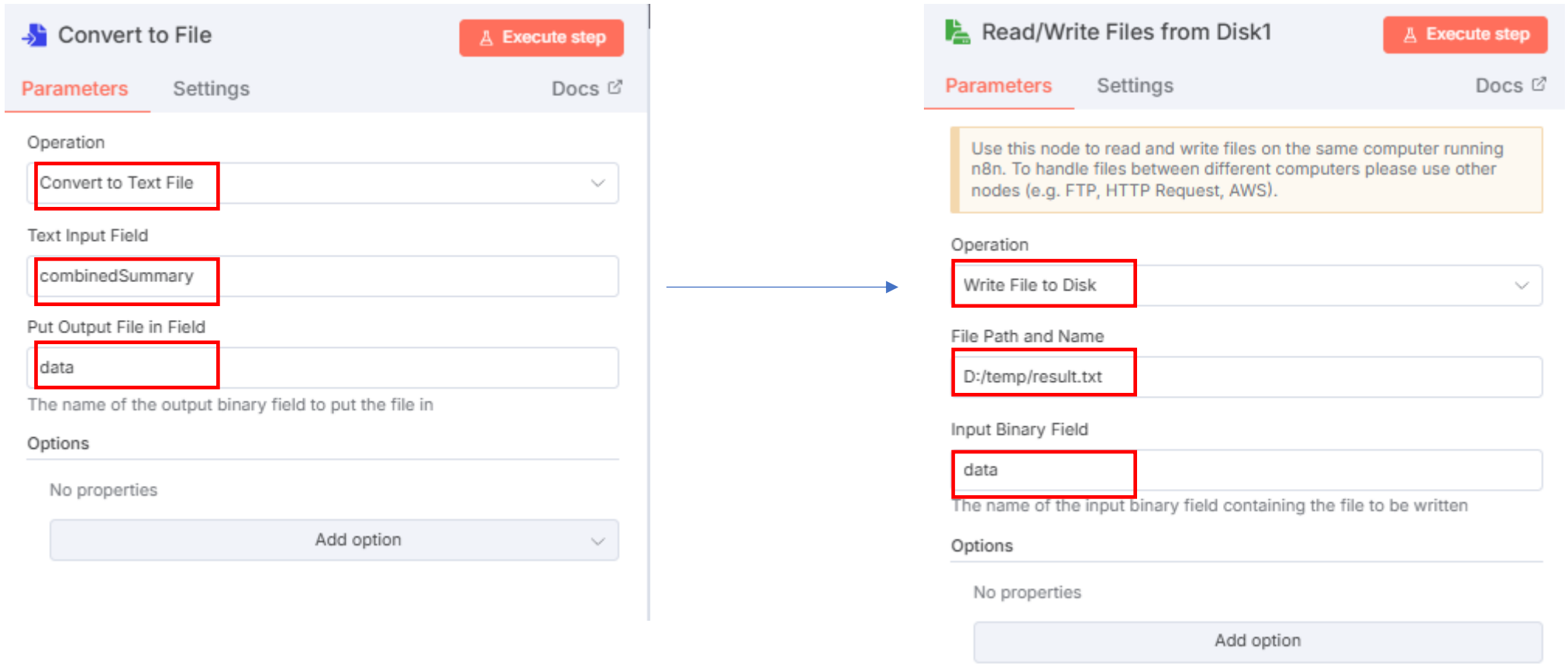
# Build the combined text with clear separation
lines = []
for index, item in enumerate(summaries, start=1):
    # Use basename if available (from Edit Fields node), otherwise fallback to generic header
    lines.append(item['output'].strip())
    lines.append("\n\n")  # Blank line between resumes

# Optional: Add a header with date/time
from datetime import datetime
header = f"RESUME BATCH SUMMARY - Processed on {datetime.now().strftime('%Y-%m-%d %H:%M:%S')}\n"
header += f"Total resumes processed: {len(summaries)}\n"
header += "=" * 50 + "\n\n"

combined_text = header + "".join(lines)

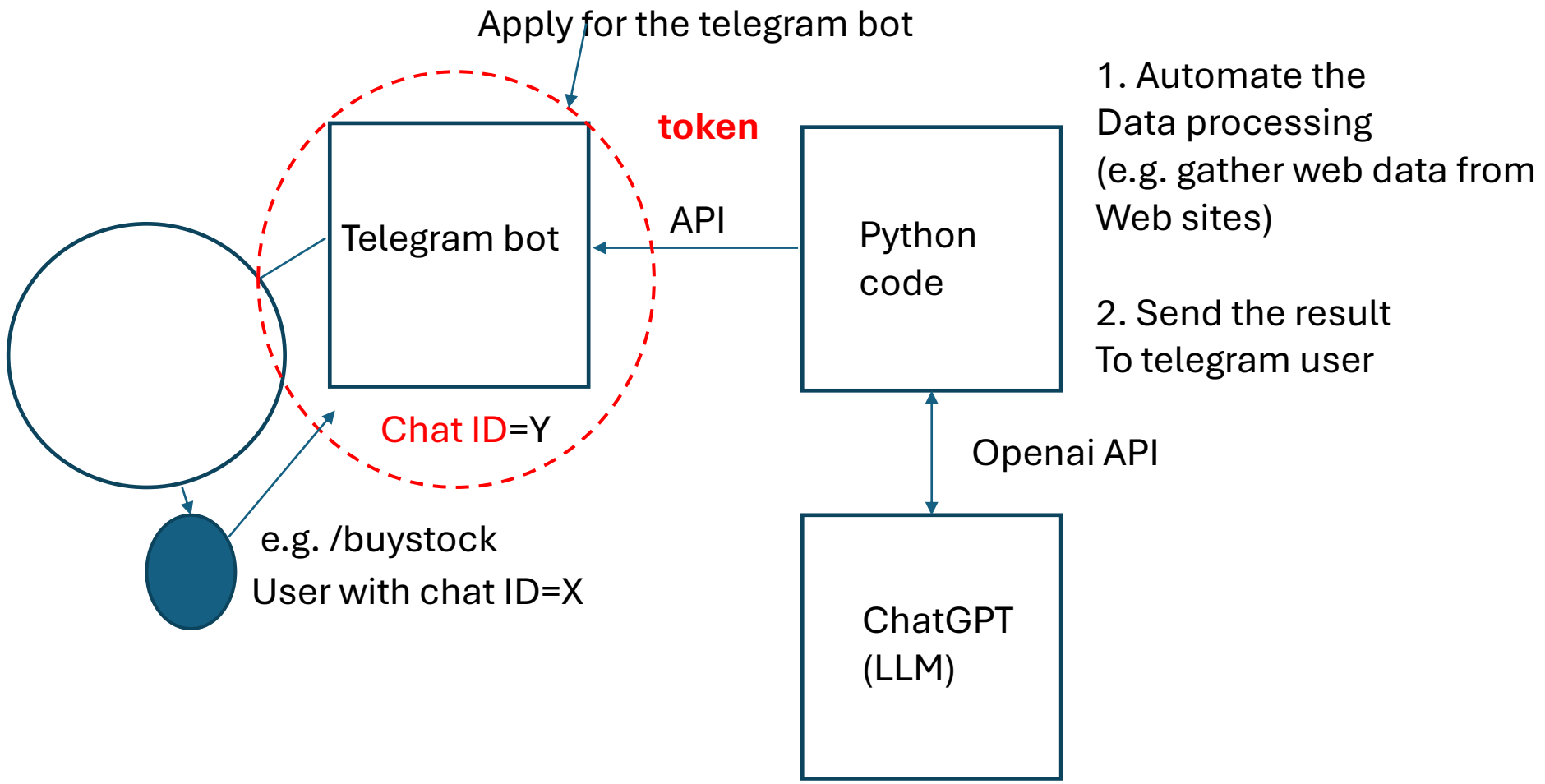
# Return the result as a new JSON field
return [{
    'json': {
        'combinedSummary': combined_text
    }
}]
```

Convert to File & Read/Write Files from Disk



Telegram API

Telegram API with Python



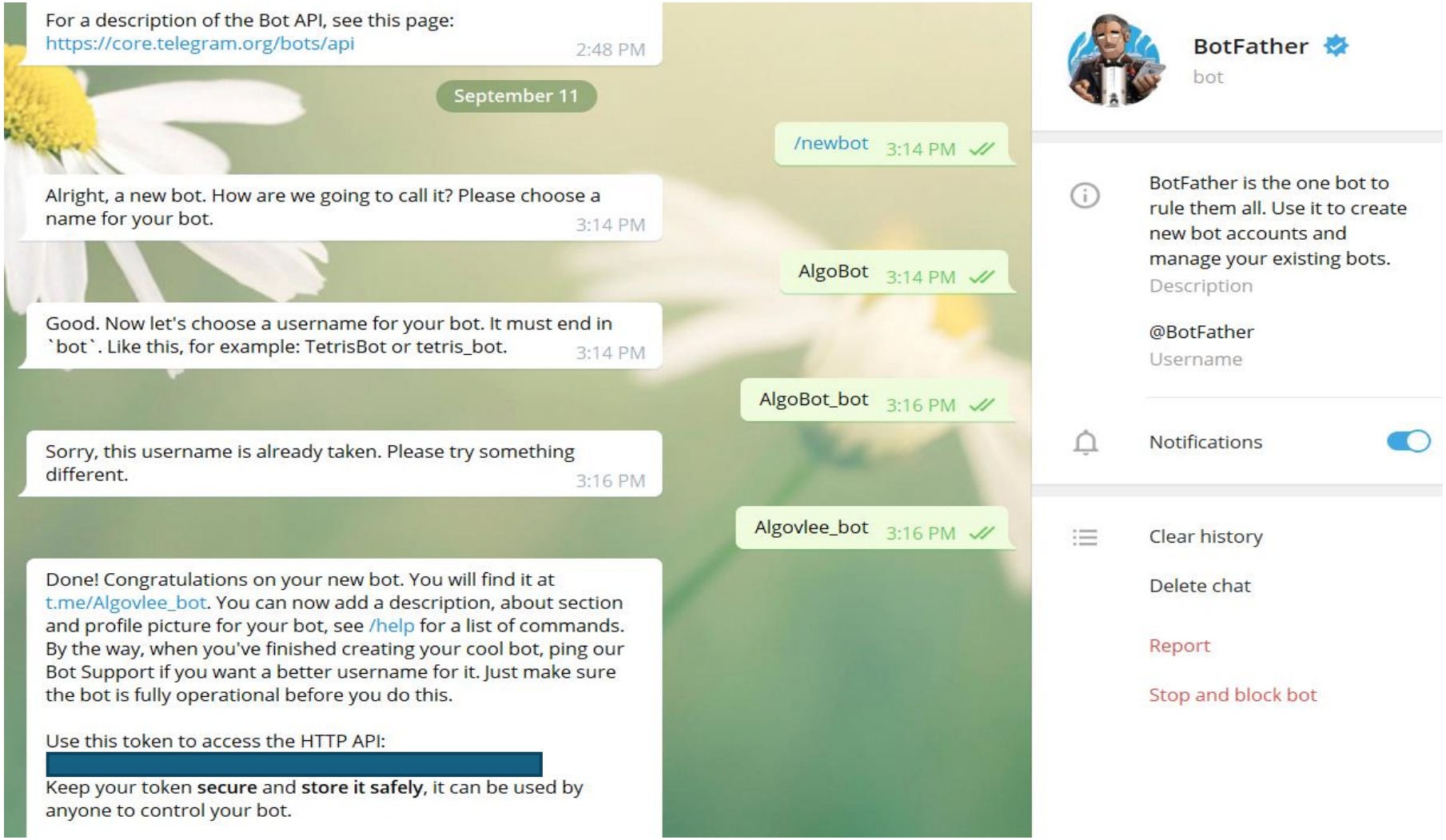
Step one

- Create Telegram account
- Login with your mobile phone
- Download Telegram desktop application

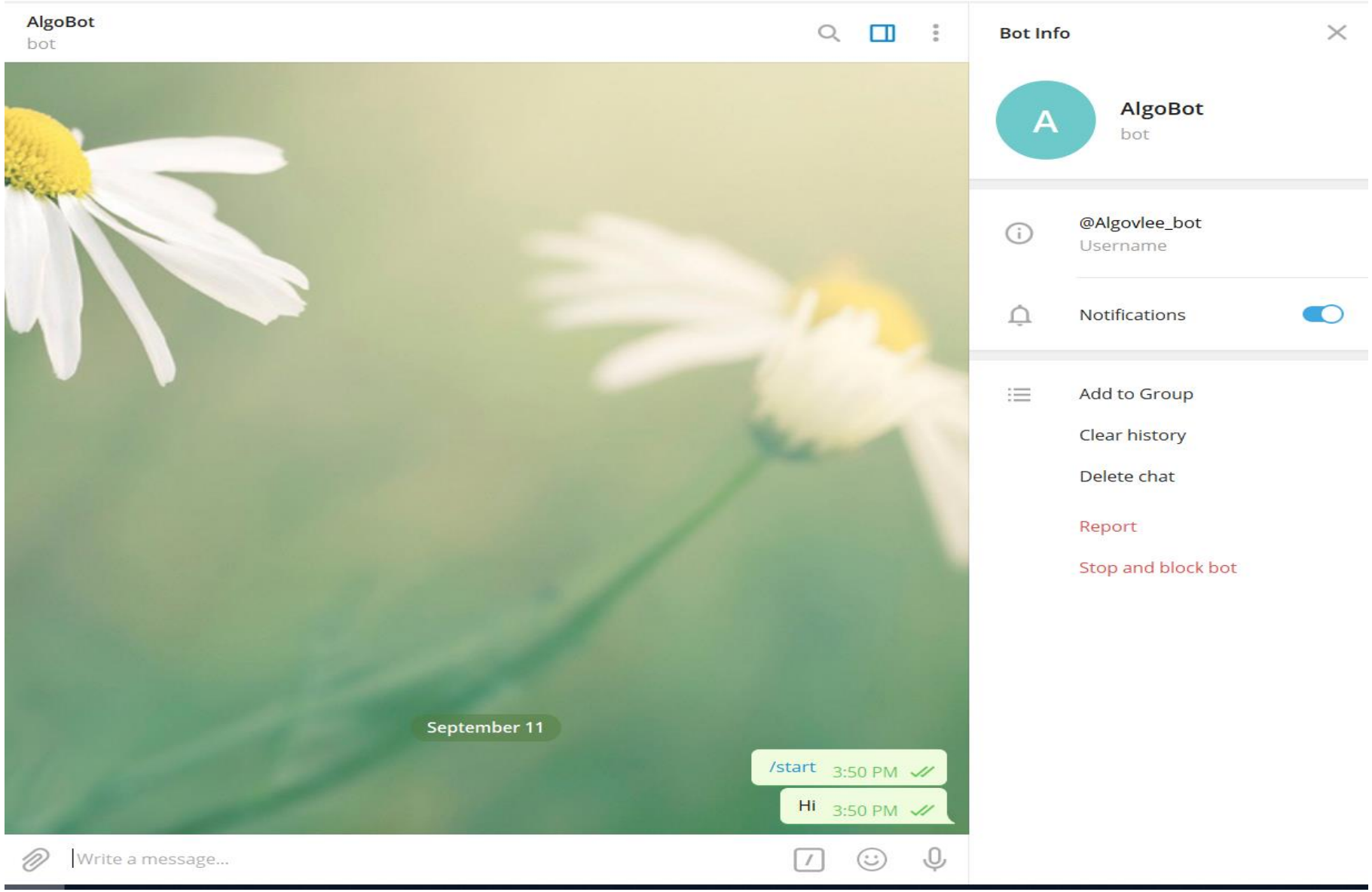
Step two

- To setup a bot in the Telegram:
- Open a chat with the **‘BotFather’** from within Telegram.
- Type in the command **/newbot**
- It will prompt you to enter a **bot name** and send you a access token.
- Open a new chat with your newly created bot.
- Click startbot
- Type something in to activate the chat.

Example


For a description of the Bot API, see this page:
https://core.telegram.org/bots/api
2:48 PM
September 11
/newbot 3:14 PM
Alright, a new bot. How are we going to call it? Please choose a name for your bot.
3:14 PM
AlgoBot 3:14 PM
Good. Now let's choose a username for your bot. It must end in `bot`. Like this, for example: TetrisBot or tetris_bot.
3:14 PM
AlgoBot_bot 3:16 PM
Sorry, this username is already taken. Please try something different.
3:16 PM
Algovlee_bot 3:16 PM
Done! Congratulations on your new bot. You will find it at t.me/Algovlee_bot. You can now add a description, about section and profile picture for your bot, see /help for a list of commands. By the way, when you've finished creating your cool bot, ping our Bot Support if you want a better username for it. Just make sure the bot is fully operational before you do this.
Use this token to access the HTTP API:
[redacted]
Keep your token secure and store it safely, it can be used by anyone to control your bot.
BotFather
bot
BotFather is the one bot to rule them all. Use it to create new bot accounts and manage your existing bots.
Description
@BotFather
Username
Notifications
Clear history
Delete chat
Report
Stop and block bot

Open a new chat with your newly created bot



Step three

- Go to the following URL – **https://api.telegram.org/botxxxxxxxxxxxxxx/getUpdates** – replacing the XXX with your access token. Here you should see a JSON structure. Note down the id (not to be confused with update_id or message_id).
- We would like to get the chat ID from here

Step three -screenshot

Your chat ID



- {"ok":true,"result":[{"update_id":XXXXXX,
- "message":{"message_id":21,"from":{"id":**XXXXX**,"is_bot":false,"first_name":"Victor Lee","username":"vickslee","language_code":"en"},"chat":{"id":XXXX,"first_name":"Victor.....

Note: You may need to enter some initial text in your chatbot and **refresh** the web link

Step four

- Go to Jupyter python
- Install package
- ! pip install python-telegram-bot
- Define your token and chat ID
- Done ☺

Use Case

Send result file to telegram bot

